

Towards Risk Indication In Mountain Biking Using Smart Wearables

Stefan Langer
stefan.langer@ifi.lmu.de
LMU Munich
Munich, Germany

Dennis Dietz
dennis.dietz@ifi.lmu.de
LMU Munich
Munich, Germany

Andreas Butz
andreas.butz@ifi.lmu.de
LMU Munich
Munich, Germany

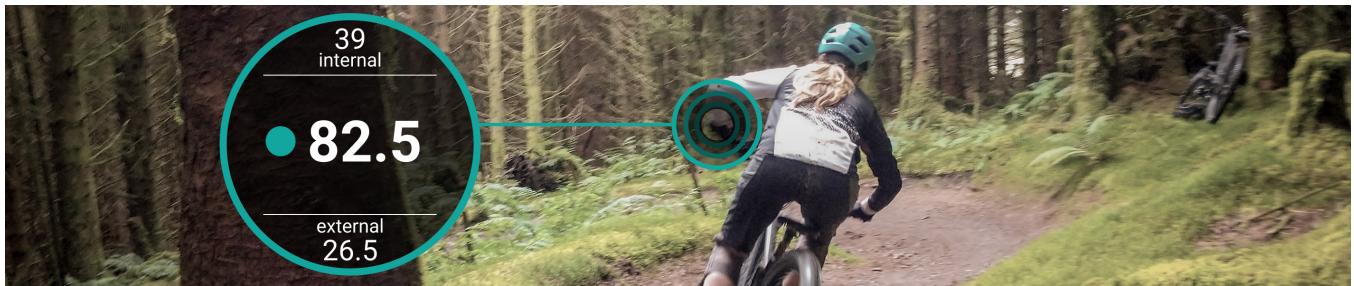


Figure 1: A watch communicates to a mountain biker their current risk of crashing, as computed from internal and external factors

ABSTRACT

Mountain biking as a recreational sport is currently thriving. During the ongoing COVID-19 pandemic, even more people started compensating for a lack of activity through individual outdoor sports, such as cycling. However, when executed beyond paved forest roads, mountain biking is a sport with subjective and objective risks, in which crashes often can not be entirely avoided and athletes may get injured. In this late-breaking work, we showcase a concept for a crash risk indication application for sports smartwatches. First, we review a wide range of related work, which formed the basis for our crash risk indication metric. We discuss options for the sensor-based detection of internal and external risk factors and propose a way to aggregate them, which will allow dynamic and potentially automatic fine-tuning by observing or obtaining feedback from the athlete. In addition, we present a concept for a smartwatch application that will provide constant feedback and an unobtrusive signal to the athlete when an unusually high risk is detected. Finally, we give an outlook on the necessary steps to implement our approach as a smartwatch app.

CCS CONCEPTS

• **Human-centered computing** → Ubiquitous and mobile computing systems and tools; Mobile devices; Haptic devices; HCI

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

CHI '2021, May 8–13, 2021, Yokohama, Japan

© 2021 Copyright held by the owner/author(s). Publication rights licensed to ACM.

ACM ISBN 978-1-4503-8095-9/21/05...\$15.00

<https://doi.org/10.1145/3411763.3451746>

theory, concepts and models; • **Applied computing** → Health informatics; Consumer health.

KEYWORDS

Smartwatch Application, Machine Learning, Risk Indication, Health, Sports, Mountain Biking

ACM Reference Format:

Stefan Langer, Dennis Dietz, and Andreas Butz. 2021. Towards Risk Indication In Mountain Biking Using Smart Wearables. In *CHI Conference on Human Factors in Computing Systems Extended Abstracts (CHI '21 Extended Abstracts)*, May 8–13, 2021, Yokohama, Japan. ACM, New York, NY, USA, 7 pages. <https://doi.org/10.1145/3411763.3451746>

1 INTRODUCTION

Many aspects of our daily lives are changing with the increasing availability of data and data-driven analysis methods. Specifically, in sports, data analytics has become a very valuable tool for optimizing training, performance, and strategy. In team sports, the analysis often focuses on player training load, player performance, or return-to-play protocols¹. The professional use of sports data has become commonplace, and companies such as Kinexon² offer sports analytics soft- and hardware. End-to-end analysis systems are primarily used in money-intensive team sports such as soccer, football, ice hockey, or basketball. While these tools focus on a risk-reward ratio, consumer-level sports and fitness trackers offer improving approximations of physical performance through, for example, speed, power, and heart rate measurements.

Companies like GARMIN³, Polar⁴, or Suunto⁵ have successfully developed fitness trackers for individual athletes for years.

¹<https://kinexon.com/uploads/images/Sports/Case-Study-St.-Johns-University.pdf>

²<https://www.kinexon.com>

³<https://www.garmin.com>

⁴<https://www.polar.com>

⁵<https://www.suunto.com>

While entry-level devices only contain IMU and heart rate sensors, more professional devices also include positioning hardware (GPS, Glonass, Galileo) and the ability to connect with accessories such as heart rate belts or power meters. State-of-the-art devices and software now provide profiles similar to those in professional end-to-end systems. For example, in the GARMIN Smartphone app, the duration, intensity, heart rate, step frequency, and other recorded data can be retrieved after training. Some sports watches even correctly identify and count swim strokes or calculate running efficiency metrics from vertical movements as measured by IMUs in the heart rate strap and the watch. Furthermore, the devices approximate training effectiveness and load and advise the athlete to take a break or reduce the load after intensive sessions. Even though these systems can help prevent injuries by training overload, they can not prevent injuries caused by unexpected events such as a crash. For individual sports like mountain biking, we believe external factors, such as track difficulty or track condition, in conjunction with the athlete's fatigue, and emotional states are strong indicators of injury probability. A mountain biker who rides on a course of suitable difficulty, reacting quickly and accurately, is likely to have a safer downhill run than a mountain biker riding on the edge of their abilities. A lack of attention, idleness, or an imbalance between skills and the level of difficulty can quickly have serious unwanted consequences. In this work, we propose the idea of a smartwatch application that uses human-in-the-loop machine learning algorithms and can indicate an estimate of the current risk of crashing to mountain bike riders.

2 BACKGROUND AND RELATED WORK

An injury in sports can have many causes. While injury prevention methods by detecting, for example, training overload exist, there is hardly any work found on the risk of crashing in sports such as mountain biking. This section goes into depth with general work on injury prevention as well as the assessment of external and internal factors that might lead to an enhanced risk of crashing. Furthermore, we take a look at existing interaction concepts in dynamic situations in order to get an idea of how to communicate a detected high risk to the athlete during an activity.

2.1 Injury Prevention

In 1992, Mechelen et al. [41] investigated the causes and effects of injuries in sports. The authors make a case for systematic injury prevention and describe the countermeasures as a *sequence of prevention*. This sequence consists of four steps, namely the evaluation of the extent of the sports injury problem, the factors influencing the injury, the identification of countermeasures, and finally, the evaluation of the measures in step one. Internal and external risk factors for sports injuries must be identified, bearing in mind that injuries can have multiple causes, including the athlete's behavior. A few years later, Meeuwisse et al. [28] proposed a multifactorial model to evaluate the causes of sports injuries. They describe the risk of injury due to the interaction of various factors and how this interaction has received little attention to date. In 2005, Bahr et al. [3] revisited multifactorial models and confirmed the need to understand the events leading to an injury. Nevertheless, the authors dispute that even 11 years after Meeuwisse's work [28],

there is still insufficient knowledge about the interaction of these causal factors. Further, Hagel et al. [17] discussed the effect of injury prevention mechanisms on risk perception. They discuss, for example, whether a helmet gives a skier a lower perceived risk and hence makes them ride faster or on more difficult terrain. Seo et al. [37] tried to classify injury risk in mountain biking; however, they did not consider sensory or physiological data nor publish their data set. Hence, their work did not provide a feasible starting point for developing a risk indication system. Verhagen et al. [43] refer to not only physiological but also behavioral factors leading to injuries. They divide injury-related behavior into preventive measures such as wearing protective equipment, sports behavior such as actions on the playing field, and rehabilitation behavior, which describes the physical ability to return to the sport. Wu et al. [45] implemented a sensor based fall risk assessment system for elderly people and validated user acceptance in a study. The participants stated that they found the fall risk estimation to be helpful and that the interface motivated them to adopt fall prevention interventions. Given the state of the literature and the lack of a clear consensus regarding the exact combination of factors leading to injuries, we, therefore, propose to approach this problem in a data-driven way. As motivated in the introduction, we specifically target mountain bikers here, but we think that substantial parts of our approach can be generalized to other sports, such as skiing or trail running.

2.2 External Risk Factors

Siebert et al. [38] use portable eye-tracking software to inspect the visual behavior of mountain bikers when they look at images of trail situations. The authors find that experts tend to focus on the trail itself while beginners scan more of the surroundings. They conclude that external parameters can distract athletes from the main task and subsequently increase their likelihood of crashing. Services such as Komoot⁶ provide an analysis of route difficulty as well as the ground surface in their sports route planning software, both may have a strong impact on the probability of crashing. The software visualizes which conditions are to be expected on which sections of the trail. Services like Openstreetmap⁷ or Trailforks⁸ provide valuable metadata for many trails, such as different difficulty scales or information on ground surfaces. Seo et al. [37] rank the difficulty of mountain bike trails, taking into consideration the slope and surface of a track through geographic map data. However, continuously fetching meta data from web services in real-time during an activity is a hardly achievable task. Furthermore, for some countries such as South Africa, difficulty ratings for trails are rarely known. This fact led Jacobs et al. [19] to classify mountain bike trail difficulty utilizing accelerometers, finding that the roughness, width, and gradient of paths greatly impact impact on their rating. Langer et al. [23] investigate the correlation between mountain bike downhill difficulty and sensor signals, trying to objectify difficulty ratings. While they achieved good accuracy in classifying trail difficulty, their model turned out to be overfitted to one rider. After finishing an activity, newer GARMIN devices compute the mountain bike specific metric *grit*, describing a ride's

⁶<https://www.komoot.com/>

⁷<https://wiki.openstreetmap.org/wiki>

⁸<https://www.trailforks.com/>

difficulty based on elevation, gradient, and rapid changes in direction⁹. Jeyapooan et al. [20] use image processing to classify the roughness of surfaces, from which the surface type and properties can be derived. While this solution could work for real-time surface detection, it still seems unfeasible for a contemporary smartwatch due to the need for camera images and a powerful processing unit. Vail et al. [39], on the other hand, detect ground surfaces from accelerometer data on a legged robot, which could also be a viable solution during sports activities.

2.3 Internal Risk Factors

Internal factors, such as fatigue and stress, can influence the athletes' injury risk during any sports that require focus, attention, and situational awareness. While physical fatigue might be detected, for example, from deteriorating movement precision or other physiological measures, work on mental fatigue or stress can be found in the affective computing field. We, therefore, examine previous work in motion assessment and affective computing.

Motion Assessment. The measurement of muscle activation (EMG), brain activity (EEG) or eye movements are often used to gain insights to internal behavior in sports-related research [4, 7, 15]. The effect of lower limb muscle activity on EMG measurements in road cycling, for example, has been studied extensively [33, 35]. Velloso et al. [42] applied a sensor- and model-based approach to detecting wrongly executed weight lifting exercises. Ladha et al. [22] assessed the skill level of sports rock climbers using an ear-worn accelerometer sensor system. Möller et al. [29] propose a smartphone-based approach to analyse exercises executed on a balance board. Roberts-Thomson et al. [31] present a fuzzy logic-based algorithm to determine jump duration, jump height and jump distance in individual sports such as skiing, snowboarding, or gymnastics. Charlton et al. [10] apply similar methods to ultimately determine the jump load of handball players and to counteract training overload. Ebert et al. [13] determined the difficulty levels of bouldering routes through IMU sensors and checked for a correct execution of fitness exercises utilizing the same hardware [12]. Burchfield et al. [8] use wireless sensor networks to detect abnormal motion sequences. Newer GARMIN devices provide the mountain bike specific metric *flow* after finishing activity recording. This metric describes how well the athlete keeps speed across slopes and through turns⁹. Concluding from their description of the metric, we believe it should not be mistaken for the affective state of flow.

Affective Computing. The computational research on emotions and affective states has become a wide research field since Picard et al. published their book on *Affective Computing* in 2000 [30]. While *arousal*, *valence*, and *motivational intensity* are the core dimensions being researched, the majority of the work seems to focus on *arousal*, since it is tied to the autonomic nervous system and can reliably be detected from physiological sensors [34]. Greene et al. [16] state that psychophysiological stress, if present over a longer period, can lead to physiological and physical symptoms, negatively affecting the athletes health. Furthermore, the American College of Sports Medicine adds that stressful life events may increase the risk

of injuries above physical and environmental factors [2]. The authors claim that stress has consistently proven to be closely tied to athletic injury risk, leading to effects like a narrowing of attention or distraction, both being critical in mentally demanding sports like mountain biking. Estimating stress from increased heart rate (HR) and changes in heart rate variability (HRV) have been found to be a viable solution [16]. The HRV signals reflect changes in the autonomic and parasympathetic nervous systems, from which stressful events can be detected [32]. Valderas et al. [40], for example, distinguish between fear and joy through HRV data. GARMIN smartwatches make use of this connection, too, and calculate the athletes stress level through HR and HRV data¹⁰. Siebert et al. [38] state, that scenarios with an increased risk (like very exposed trails) lead to a higher arousal. This could lead to a distraction from the main task, thus enhance the likelihood of crashing. Maier et al. [26] detect the state of *flow* from physiological signals measured using wearable devices. The flow state occurs when humans are facing a challenge of optimal difficulty. We believe that being *in flow* is a potential factor for lower risk. Building on the knowledge gained by Maier et al. [26], their coworkers Bartl et al. [5] aimed to measure the flow state in biathletes' physiological data and were able to show that physiological patterns help to predict shooting performance.

2.4 Interaction and Visualization in Sports and Rehabilitation

In a survey from 2009, Alahakone et al. [1] summarize interaction possibilities in sports, rehabilitation and gaming with vibrotactile feedback systems. In rehabilitation, vibrotactile feedback is often used to correct mistakes in movement or posture. For example, the "TactaPack" [25] supports patients who received an artificial joint by providing vibrotactile feedback for the correct execution of movements. The same method is used to counteract knee hyperextension or instability of patients with balance disorders. Jacobs [18] tested tactile feedback patterns to give snowboarders feedback on their riding posture during rides and achieved a high accuracy in understanding information from vibrational feedback. In addition to tactile feedback, also auditory feedback has been considered an effective medium for interaction during sports. Cesarini et al. [9] developed a system that measures the displaced water mass during swimming. This value is transmitted to the athlete via audio signals and has proven to help optimizing the swimming technique with high acceptance among study participants. Feeken et al. [14] integrated pressure sensors into climbing shoes and tested the effectiveness of auditory feedback during climbing. While audio signals can only be poorly perceived in a noisy environment, such as a climbing gym, they might be more feasible in quieter environments. Baudry et al. [6] conducted a study in which they measured a positive effect of auditory feedback on their body segmental alignment during pommel horse exercises. While also visual feedback has been researched in sports and rehabilitation environments [11, 21, 24, 44], we believe that it will create visual distraction, hence increase the risk of crashing.

⁹<https://support.garmin.com/en-US/?faq=LmMrHllquc8loZMhtxZ0i7>

¹⁰<https://www8.garmin.com/manuals/webhelp/fenix5/EN-US/GUID-20E035CE-CA52-40D0-B202-13403D6FAA51.html>

3 A METRIC FOR INJURY RISK

To be able to communicate *risk* to the athlete, we first need a reliable risk estimation metric. From the related work in [section 2](#), we hypothesize that a high amount of stress and exhaustion increases the likelihood of crashing in mountain biking and that a high difficulty of the trail further magnifies this risk. Moreover, we believe that an adequate skill-to-difficulty ratio has a positive influence on the athlete's safety. By aggregating the individual risk components, we can quantify the level of risk. However, due to uncertainty of how strongly each metric influences the overall risk, we add weight factors that adjust whenever the user gives reinforcement. When, for example, an athlete labels a ride as having been of high risk, metrics that strongly deviate from their median value can be weighted heavier in future calculations.

$$r = \sum_{i=1}^n [w_i * (\frac{m_i - m_i^{min}}{m_i^{max} - m_i^{min}})^k] \quad (1)$$

[Equation 1](#) shows how we can calculate an aggregated risk metric, with n being the number of individual risk factors, w_i being the weight for each sub-metric m_i and k a constant exponent. We propose to initialize all weights w_i with a value of $1/n$ which means that they will initially give equal influence to all factors and add up to 1. Hence the overall risk will be guaranteed to be between 0 and 1. For $k = 1$, this will simply compute the arithmetic mean, and for $k = 2$ the mean square. The formula also normalizes sub-metrics to a range of 0 to 1 by considering their m_i^{min} and m_i^{max} values, for example, across all activities recorded by the athlete before. The weights can subsequently be fine-tuned automatically from observations or by the user's feedback through the smartwatch application. Values of $0 < k < 1$ will amplify the influence of small values of m_i , while $k > 1$ will increasingly emphasize only larger values. For better readability, the resulting risk should probably be displayed as a percentage by multiplying r with 100.

[Equation 1](#) represents the first proposal and assumes that the resulting risk can reasonably be calculated as a linear combination of factors. It has the advantage that it will work well with other established methods, e.g., a multifactorial linear regression for making risk predictions. However, there is a chance that in practical trials, we will find the metric to be overly simplified and that some individual factors might have a non-linear influence or might even better be represented by multiplying their individual values. To make this more concrete, riding on a flat, paved road will make the risk of a crash practically zero, independent of all other factors, such as fatigue or ability. We hence propose [Equation 1](#) as a working model to start with but are prepared to readjust this later in our research.

3.1 External Risk Indicators

The **Difficulty** of a mountain bike downhill track has a strong influence on the selection of the track [\[23\]](#). While novice riders are likely to ride mainly easy trails, facing a lower risk of crashing, more advanced riders will presumably often pick more challenging paths. We hypothesize that a high difficulty implies a higher risk of crashing. However, quantifying difficulty is not a trivial task, leading to subjective ratings that differ from area to area. Research has shown that difficulty can be estimated from physiological and

IMU sensor data [\[19, 23, 37\]](#). We, therefore, propose to circumvent the problem of subjective trail ratings by computing the difficulty from physical and physiological sensor data provided by the smartwatch. Most importantly, we believe that the **Surface** of the track influences the difficulty rating most. While this information is likely implied in the data used within the difficulty metric, we propose to give it more weight by classifying the surface through acceleration sensor data as it's own sub-metric.

3.2 Internal Risk Indicators

Stress is directly tied to an athlete's injury risk [\[2\]](#). We have learned that chronic stress or stressful life events can distract from the actual task, lower attention, and enlarge the crashing risk. We can approximate this long-lasting stress level from HRV data [\[16\]](#), which has to be recorded and analyzed before the activity starts. Fortunately, smart sports wearables like the *GARMIN Fenix* watch¹¹ readily compute and provide this metric throughout the day. Nevertheless, acute stress can also distract from the task at hand and has to be computed during the activity. Novice riders in scary riding scenarios, for example, tend to be distracted from the actual trail, scanning their surroundings [\[38\]](#). We can approximate this kind of acute stress (fear) from spectral-domain indices in HRV data [\[40\]](#). However, this approach's accuracy first has to be validated in dynamic scenarios such as mountain biking. Besides stress, we believe **Fatigue** to have a large influence on the risk of crashing. GARMIN provides a metric for the athlete's performance condition, claiming that it can be used as an indicator of current fatigue, taking into account speed, HR, HRV, and power (if additional hardware is present). This metric ranges from -20 to +20 and will be recorded during an activity, starting in the first 6 to 20 minutes. Approximating fatigue from HRV has been researched extensively [\[27, 36\]](#) and found to be a viable solution for our use case. In addition, some devices also compute an expected time until full recovery after each workout¹². If this timer indicates that the athlete has not yet fully recovered from previous training, this is an indicator of latent fatigue, even without continuous measurements.

3.3 Mixed Risk Indicators

In psychology and affective computing research, the *flow* state is described as something that most likely occurs when the athlete faces a difficulty that is neither too easy nor too challenging [\[5, 26\]](#). From a flow metric, we can derive the athlete's feeling of appropriate **skill-to-difficulty ratio**, which allows us to estimate a lower crash risk. However, further research is needed for mountain biking on how to recognize flow from physical and physiological sensor data.

4 ADEQUATE RISK INDICATION

To achieve adequate risk indication, we need to answer the questions "When do we indicate high risk?" and "How do we indicate high risk?". To do so, we developed a concept for the smartwatch application as well as for the indication of a high-risk estimate.

¹¹<https://discover.garmin.com/en-US/fenix-6/>

¹²<https://www.firstbeatanalytics.com/en/features/recovery-time/>

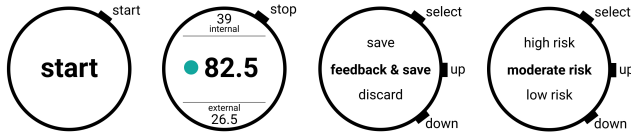


Figure 2: The four key steps in the interaction concept for our proposed risk indicator application: The athlete initially starts the recording of the activity (left). During the activity (center left), the watch displays a risk value between 0 and 100, potentially further detailed into internal and external risk factors, and with an optional flow state indicator (colored dot). When the activity is stopped (centre right & right), the rider can either save or discard the recording and provide feedback on the risk level they sensed subjectively.

4.1 Application Interaction Concept

Figure 2 shows a user interface concept of the proposed smartwatch application. Figure 2left depicts the state before starting the recording, allowing the user to start their activity similarly to starting a regular workout recording on many sports watches. The second image shows the continuous risk feedback during activity tracking. On the top, the aggregated metric for internal risk factors is shown, indicating the user’s current stress and fatigue level. On the bottom of the screen, the metric for external risk factors is shown, namely an aggregated value for difficulty and surface. In the center of the screen, the accumulated risk indication metric is displayed in a large font to preserve readability under difficult conditions. A colored dot, like an on/off light, visualizes whether a positive flow state has been detected. Screen three shows the options after stopping the activity recording. The user can either discard the recording, just save it, or save the recording after providing a subjective difficulty rating for the activity. The latter process is visualized in screen 4, showing a coarse rating scale with options for *low risk*, *moderate risk*, and *high risk*. This selection will influence the weights of the risk calculation and will be an indication for future activities.

When thinking about an even deeper integration, our calculated risk metric could be a custom data field in these watches, and athletes could determine themselves when, and in which position of the screen they want to see this. However, reading a small screen during challenging activities presents a problem in itself, so we propose to start from a rather bold layout, as shown in Figure 2.

4.2 Risk Indication Concept

The indication and interpretation of risk represents an extremely sensitive task. The user must not be given any false impression of safety, so they will not take higher risks because of it. Therefore, the metric itself is not referring to a scale or the like but describes the sum of sub-metrics. However, with time, the user can learn to interpret this metric personally as low or high risk and develop a sense for their adequate risk level.

Nevertheless, we propose to indicate an untypically high risk to mountain bikers during the activity, as depicted in Figure 3. In order to answer the question “When do we indicate a high risk?” we need a threshold above which an indication is given. This threshold can be computed in an upper n th-percentile style and further fine-tuned

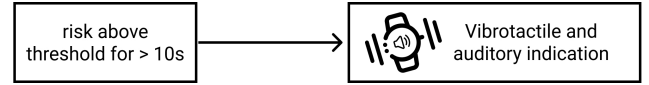


Figure 3: A risk level, that persistently exceeds a certain threshold will be indicated by additional modalities, such as vibration, sound or flashing of the screen.

through user feedback as described in subsection 4.1. A threshold should only be calculated after several recorded activities (to reduce notifications in lower-risk scenarios) or after recording at least one activity that was marked as being of high risk by the user. The user will be notified whenever their risk indication metric is above this dynamic threshold for more than a certain time, e.g., 10s. A new indication should only be given if the user was either below the threshold again for a certain time or above the threshold for a long time¹³. The application’s goal is to prompt the rider to re-evaluate the current situation and potentially lower their current risk by, for example, slowing down or taking a reroute.

Vibrotactile feedback is a viable and unobtrusive interaction concept in sports [1, 18, 25]. However, in activities such as mountain biking, many vibrations and shocks are submitted to the hands and arms, potentially drowning the risk indication signal. Therefore, we plan to extend the vibrotactile feedback by auditory feedback and strong visual indications such as screen flashing, giving an answer to the question of “How do we indicate high risk?”. Vibrotactile and auditory feedback is a typical interaction concept for smartwatches. Hence, we believe that those signals will not distract athletes from their actual task more than their usual activity tracking.

5 DISCUSSION AND FUTURE WORK

We highlighted a lot of work in the fields of stress and fatigue detection from HRV. However, external and mountain bike-specific risk indicators have not been researched much to date. For difficulty detection, experiments have shown its general feasibility [19, 23, 37]. However, a comprehensive and generic approach is still missing. For surface detection in mountain biking utilizing smartwatches, there is, to the best of our knowledge, no existing work. Hence, we first have to collect a labeled, large, and diverse data set recorded on actual trails to implement the proposed ideas in a prototype. Subsequently, we can train machine learning algorithms, which hopefully reliably classify the difficulty and surface of a track. Moreover, we are not aware of any work on detecting flow through wearable devices and sensors outside a lab environment. While an indication on the possibility of detecting flow in sports through physiological data has been given [5], more work has to be done to enable a robust indication in mountain biking. One could, for example, think of recording a professional rider during a race and then manually label states of flow in the aftermath, following the classification approach used in [26].

By providing mountain bikers a risk indication metric and high-risk alerts, we believe to be able to reduce the likelihood of crashing. Wu et al. [45] have shown that elderly people are motivated to adopt fall prevention tips given by software when a high falling risk has

¹³Concrete numerical values for these timeouts will have to be tested in prototype studies.

been indicated. As far as this can be applied to sports like mountain biking, we hope to motivate riders to walk their bike or take a reroute in risky scenarios.

6 SUMMARY AND OUTLOOK

In this work, we proposed a concept for a smartwatch application that gives mountain bikers continuous feedback about the calculated risk of their current activity and allows them to be notified of particularly risky scenarios. For this purpose, related work was evaluated to demonstrate the possibilities of a risk indication metric. In addition, an interaction concept of the application was presented. At last, we gave an outlook on the necessary steps and open challenges for implementing the approach in a prototype. Some parts of the puzzle are still missing to develop a comprehensive and reliable risk indication application for wearable devices. However, several separate proofs of concept have shown that those parts can be filled through of data and training of appropriate machine learning algorithms. Therefore, we are optimistic and will continue our research towards such a risk indication system that can help reduce the risk of crashing or injury in mountain biking and potentially other sports.

REFERENCES

- [1] Aruni Upeksha Alahakone and SMN Arosha Senanayake. 2009. Vibrotactile feedback systems: Current trends in rehabilitation, sports and information display. In *IEEE/ASME International Conference on Advanced Intelligent Mechatronics (AIM '06)*. IEEE, 1148–1153. <https://doi.org/10.1109/AIM.2009.5229741>
- [2] American College of Sports Medicine, American Academy of Family Physicians, American Academy of Orthopaedic Surgeons, American Medical Society for Sports Medicine, American Orthopaedic Society for Sports Medicine, American Osteopathic Academy of Sports Medicine, et al. 2006. Psychological issues related to injury in athletes and the team physician: a consensus statement. *Medicine and Science in Sports and Exercise* 38, 11 (2006), 2030. <https://doi.org/10.1249/mss.0b013e31802b37a6>
- [3] Roald Bahr and Tron Krosshaug. 2005. Understanding injury mechanisms: a key component of preventing injuries in sport. *British journal of sports medicine* 39, 6 (2005), 324–329. <https://doi.org/10.1136/bjsm.2005.018341>
- [4] SN Baker, Etienne Olivier, and RN Lemon. 1997. Coherent oscillations in monkey motor cortex and hand muscle EMG show task-dependent modulation. *The Journal of physiology* 501, Pt 1 (1997), 225. <https://doi.org/10.1111/j.1469-7793.1997.225bo.x>
- [5] Michael Bartl and Johann Füller. 2020. The Rise of Emotion AI: Decoding Flow Experiences in Sports. In *21st Century Sports*. Springer, 219–229.
- [6] Ludovic Baudry, David Leroy, Régis Thouvenecq, and Didier Chollet. 2006. Auditory concurrent feedback benefits on the circle performed in gymnastics. *Journal of sports sciences* 24, 2 (2006), 149–156. <https://doi.org/10.1080/02640410500130979>
- [7] Craig E Boettcher, Karen A Ginn, and Ian Cathers. 2008. Standard maximum isometric voluntary contraction tests for normalizing shoulder muscle EMG. *Journal of orthopaedic research* 26, 12 (2008), 1591–1597. <https://doi.org/10.1002/jor.20675>
- [8] T. Ryan Burchfield and Subbarayan Venkatesan. 2007. Accelerometer-Based Human Abnormal Movement Detection in Wireless Sensor Networks. In *Proceedings of the 1st ACM SIGMOBILE International Workshop on Systems and Networking Support for Healthcare and Assisted Living Environments* (San Juan, Puerto Rico) (*HealthNet '07*). Association for Computing Machinery, New York, NY, USA, 67–69. <https://doi.org/10.1145/1248054.1248073>
- [9] Daniel Cesarini, Davide Calvaresi, Chiara Farnesi, Diego Taddei, Stefano Frediani, Bodo Ungerechts, and Thomas Hermann. 2016. MEDIATION: an eMbEddeD system for auditory feedback of hand-water InterAction while swimming. *Procedia engineering* 147 (2016). <https://doi.org/10.1016/j.proeng.2016.06.301>
- [10] Paula C Charlton, Claire Kenneally-Dabrowski, Jeremy Sheppard, and Wayne Spratford. 2017. A simple method for quantifying jump loads in volleyball athletes. *Journal of science and medicine in sport* 20, 3 (2017), 241–245. <https://doi.org/10.1016/j.jsams.2016.07.007>
- [11] Harrison Philip Crowell, Clare E Milner, Joseph Hamill, and Irene S Davis. 2010. Reducing impact loading during running with the use of real-time visual feedback. *Journal of orthopaedic & sports physical therapy* 40, 4 (2010), 206–213. <https://doi.org/10.2519/jospt.2010.3166>
- [12] André Ebert, Marie Kiermeier, Chadly Marouane, and Claudia Linnhoff-Popien. 2017. Sensx: About sensing and assessment of complex human motion. In *2017 IEEE 14th International Conference on Networking, Sensing and Control (ICNSC '18)*. IEEE, 327–332. <https://doi.org/10.1109/ICNSC.2017.8000113>
- [13] André Ebert, Kyrill Schmid, Chadly Marouane, and Claudia Linnhoff-Popien. 2017. Automated recognition and difficulty assessment of boulder routes. In *International Conference on IoT Technologies for HealthCare*. Springer, 62–68. https://doi.org/10.1007/978-3-319-76213-5_9
- [14] Corinna Feeken, Merlin Wasmann, Wilko Heuten, Dag Ennenga, Heiko Müller, and Susanne Boll. 2016. ClimbingAssist: direct vibro-tactile feedback on climbing technique. In *Proceedings of the 2016 ACM International Joint Conference on Pervasive and Ubiquitous Computing: Adjunct*. 57–60. <https://doi.org/10.1145/2968219.2971417>
- [15] R Ferber, LR Osternig, and DC Gravelle. 2002. Effect of PNF stretch techniques on knee flexor muscle EMG activity in older adults. *Journal of electromyography and kinesiology* 12, 5 (2002), 391–397. [https://doi.org/10.1016/S1050-6411\(02\)00047-0](https://doi.org/10.1016/S1050-6411(02)00047-0)
- [16] Shalom Greene, Himanshu Thapliyal, and Allison Caban-Holt. 2016. A survey of affective computing for stress detection: Evaluating technologies in stress detection for better health. *IEEE Consumer Electronics Magazine* 5, 4 (2016), 44–56. <https://doi.org/10.1109/MCE.2016.2590178>
- [17] Brent Hagel and Willem Meeuwisse. 2004. Risk compensation: a “side effect” of sport injury prevention? <https://doi.org/10.1097/00042752-200407000-00001>
- [18] Mareike Jacobs. 2008. Design and recognition of tactile feedback patterns for snowboarding. *Computer Science Department RWTH Aachen University* (2008).
- [19] Wietsche M.G. Jacobs, Wynand J. Steyn, and Handre Van Niekerk. 2014. Classification of mountain bike trails using vehicle-pavement interaction principles. *South African Journal for Research in Sport, Physical Education and Recreation* 36, 1 (2014), 211–227.
- [20] Thangasamy Jeyapoovan and M. Murugan. 2013. Surface roughness classification using image processing. *Measurement* 46, 7 (2013), 2065–2072. <https://doi.org/10.1016/j.measurement.2013.03.014>
- [21] Eleftherios Kellis and Vasilios Baltzopoulos. 1996. Resistive eccentric exercise: effects of visual feedback on maximum moment of knee extensors and flexors. *Journal of Orthopaedic & Sports Physical Therapy* 23, 2 (1996), 120–124. <https://doi.org/10.2519/jospt.1996.23.2.120>
- [22] Cassim Ladha, Nils Y Hammerla, Patrick Olivier, and Thomas Plötz. 2013. ClimbAX: skill assessment for climbing enthusiasts. In *Proceedings of the 2013 ACM international joint conference on Pervasive and ubiquitous computing*. 235–244. <https://doi.org/10.1145/2493432.2493492>
- [23] Stefan Langer, Robert Müller, Kyrill Schmid, and Claudia Linnhoff-Popien. 2019. Difficulty Classification of Mountainbike Downhill Trails utilizing Deep Neural Networks. In *Joint European Conference on Machine Learning and Knowledge Discovery in Databases*. Springer, 270–280. https://doi.org/10.1007/978-3-030-43887-6_21
- [24] Dario G Lieberman, Larry Katz, Mike D Hughes, Roger M Bartlett, Jim McClements, and Ian M Franks. 2002. Advances in the application of information technology to sport performance. *Journal of sports sciences* 20, 10 (2002), 755–769. <https://doi.org/10.1080/02640410220675611>
- [25] Robert W Lindeman, Yasuyuki Yanagida, Kenichi Hosaka, and Shinji Abe. 2006. The TactaPack: A wireless sensor/actuator package for physical therapy applications. In *14th Symposium on Haptic Interfaces for Virtual Environment and Teleoperator Systems (HAPTIC '06)*. IEEE, 337–341. <https://doi.org/10.1109/HAPTIC.2006.1627117>
- [26] Marco Maier, Daniel Elsner, Chadly Marouane, Meike Zehnle, and Christoph Fuchs. 2019. DeepFlow: Detecting Optimal User Experience From Physiological Data Using Deep Neural Networks. In *Proceedings of the Twenty-Eighth International Joint Conference on Artificial Intelligence (IJCAI '19)*. International Joint Conferences on Artificial Intelligence Organization, 1415–1421. <https://doi.org/10.24963/ijcai.2019/196>
- [27] Bojan Makivić, Marina Nikić Djordjević, and Monte S Willis. 2013. Heart Rate Variability (HRV) as a Tool for Diagnostic and Monitoring Performance in Sport and Physical Activities. *Journal of Exercise Physiology Online* 16, 3 (2013).
- [28] Willem H. Meeuwisse. 1995. Assessing Causation in Sport Injury. *Clinical Journal of Sport Medicine* 4 (1995), 166–170.
- [29] Andreas Möller, Luis Roalter, Stefan Diewald, Johannes Scherr, Matthias Kranz, Nils Hammerla, Patrick Olivier, and Thomas Plötz. 2012. GymSkill: A personal trainer for physical exercises. In *2012 IEEE International Conference on Pervasive Computing and Communications (PerCom '12)*. IEEE, 213–220. <https://doi.org/10.1109/PerCom.2012.6199869>
- [30] Rosalind W. Picard. 2000. *Affective computing*. MIT press.
- [31] Claire L Roberts-Thomson, Anatole M Lokshin, and Vitaly A Kuzkin. 2014. Jump detection using fuzzy logic. In *2014 IEEE Symposium on Computational Intelligence for Engineering Solutions (CIES '14)*. IEEE, 125–131. <https://doi.org/10.1109/CIES.2014.7011841>
- [32] Brian F Robinson, Stephen E Epstein, G David Beiser, and Eugene Braunwald. 1966. Control of heart rate by the autonomic nervous system: studies in man on the interrelation between baroreceptor mechanisms and exercise. *Circulation Research* 19, 2 (1966), 400–411. <https://doi.org/10.1161/01.RES.19.2.400>

- [33] David M Rouffet and Christophe A Hautier. 2008. EMG normalization to study muscle activation in cycling. *Journal of Electromyography and Kinesiology* 18, 5 (2008), 866–878. <https://doi.org/10.1016/j.jelekin.2007.03.008>
- [34] Stanisław Saganowski, Anna Dutkowiak, Adam Dziadek, Maciej Dzieżyc, Joanna Komoszyńska, Weronika Michalska, Adam Polak, Michał Ujma, and Przemysław Kazienko. 2020. Emotion recognition using wearables: A systematic literature review–work-in-progress. In *2020 IEEE International Conference on Pervasive Computing and Communications Workshops (PerCom Workshops '20)*. IEEE, 1–6. <https://doi.org/10.1109/PerComWorkshops48775.2020.9156096>
- [35] Gerrit Jan Van Ingen Schenau. 1989. From rotation to translation: Constraints on multi-joint movements and the unique action of bi-articular muscles. *Human Movement Science* 8, 4 (1989), 301–337. [https://doi.org/10.1016/0167-9457\(89\)90037-7](https://doi.org/10.1016/0167-9457(89)90037-7)
- [36] Laurent Schmitt, Jacques Regnard, and Grégoire P Millet. 2015. Monitoring fatigue status with HRV measures in elite athletes: an avenue beyond RMSSD? *Frontiers in physiology* 6 (2015), 343. <https://doi.org/10.3389/fphys.2015.00343>
- [37] Dong-Hyok Seo, Suk Ho Jung, Soo-Ho Park, Kwang-Sun Ryu, Hyeonsoo Kim, Dong Gyu Lee, and Keun Ho Ryu. 2011. Assessment building a method for risk model of mountain bike accident based on classification techniques. In *International Conference on Hybrid Information Technology*. Springer, 716–723. https://doi.org/10.1007/978-3-642-24082-9_87
- [38] Stefan Siebert and M Kolleck. 2013. Difficulty rating in mountain biking. *Journal of outdoor activities* 7 (2013), 88–93.
- [39] Douglas Vail and Manuela Veloso. 2004. Learning from accelerometer data on a legged robot. In *Proceedings of the 5th IFAC/EURON symposium on intelligent autonomous vehicles*. Elsevier. [https://doi.org/10.1016/S1474-6670\(17\)32082-7](https://doi.org/10.1016/S1474-6670(17)32082-7)
- [40] Maria Teresa Valderas, Juan Bolea, Pablo Laguna, Montserrat Vallverdú, and Raquel Bailón. 2015. Human emotion recognition using heart rate variability analysis with spectral bands based on respiration. In *37th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC '15)*. IEEE, 6134–6137. <https://doi.org/10.1109/EMBC.2015.7319792>
- [41] Willem Van Mechelen, Hynek Hlobil, and Han C. G. Kemper. 1992. Incidence, severity, aetiology and prevention of sports injuries. *Sports medicine* 14, 2 (1992), 82–99. <https://doi.org/10.2165/00007256-199214020-00002>
- [42] Eduardo Velloso, Andreas Bulling, Hans Gellersen, Wallace Ugulino, and Hugo Fuks. 2013. Qualitative activity recognition of weight lifting exercises. In *Proceedings of the 4th Augmented Human International Conference*. 116–123. <https://doi.org/10.1145/2459236.2459256>
- [43] Evert ALM Verhagen, Maartje M van Stralen, and Willem Van Mechelen. 2010. Behaviour, the key factor for sports injury prevention. *Sports medicine* 40, 11 (2010), 899–906. <https://doi.org/10.2165/11536890-000000000-00000>
- [44] Catherine Walker, Brenda J Brouwer, and Elsie G Culham. 2000. Use of visual feedback in retraining balance following acute stroke. *Physical therapy* 80, 9 (2000), 886–895. <https://doi.org/10.1093/ptj/80.9.886>
- [45] Alan Yusheng Wu and Cosmin Munteanu. 2018. Understanding older users' acceptance of wearable interfaces for sensor-based fall risk assessment. In *Proceedings of the 2018 CHI conference on human factors in computing systems*. 1–13. <https://doi.org/10.1145/3173574.3173693>